

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Zoology**Paper: ZOB-201 — Fundamental of Cell Biology and Elementary Biochemistry**

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A (including Question No. 1 which is compulsory) and three questions from Section-B (including Question No. 1 which is compulsory). Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A**(Fundamental of Cell Biology) (Marks: 35)****Question 1.** Answer the following parts:

- (a) Select and write the best options in the following: $[4 \times \frac{1}{2} = 2]$
- (i) The tails of the phospholipids of the plasma membrane are composed of _____ and _____.
- (1) phosphate groups, hydrophobic
 - (2) fatty acid groups, hydrophilic
 - (3) phosphate groups, hydrophilic
 - (4) fatty acid groups, hydrophobic
- (ii) Which of the following do not participate in innate immunity?
- (1) Natural killer cells
 - (2) Phagocytic cells
 - (3) Polymorphonuclear leukocytes
 - (4) T cells
- (iii) After meiosis-I daughter cells will have:
- (1) twice as much DNA as a haploid gamete
 - (2) as much as a haploid gamete
 - (3) four times as much as a haploid gamete
 - (4) None of the above
- (iv) Resolving power of a microscope is a function of:
- (1) wavelength of light used
 - (2) numerical aperture lens system

- (3) refractive index
- (4) wavelength of light used and numerical aperture of lens system
- (b) State whether the following statements are True or False with brief justification: $[3 \times 2 = 6]$
 - (i) Majority of human cancers prevalent worldwide are caused due to viruses.
 - (ii) Polytene chromosomes are found in amphibian oocytes and are giant meiotic prophase chromosomes.
 - (iii) Glycosylation of lipids and proteins takes place in endosomes and nucleus, respectively.
- (c) Define the following: $[3 \times 1 = 3]$
 - (i) RB gene
 - (ii) Rhopheocytosis
 - (iii) NOR.

Question 2. Answer the following:

- (a) Describe the basic structure of immunoglobulin with suitable diagram. Discuss the characteristic features and functions of different types of immunoglobulins. [6]
- (b) Explain the role of Golgi complex in sorting of lysosomal enzymes. [6]

Question 3. Answer the following:

- (a) Explain the various steps of the ribosome biogenesis in detail using well-labelled flow chart diagram. [6]
- (b) What is the role of the inner mitochondrial membrane in ATP synthesis? [6]

Question 4. Write detailed notes on the following: [3 × 4 = 12]

- (a) Synaptonemal complex and its role in chromosome pairing
- (b) Process of transport across plasma membrane
- (c) Lampbrush chromosomes and their functions.

SECTION – B
(Elementary Biochemistry) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False and give brief justification in support of your answer: [3 × 2 = 6]
 - (i) Cerebrosides are derivatives of sphingolipids.
 - (ii) Shine-Dalgarno sequence helps in DNA-replication.
 - (iii) DNA Pol activity is primer dependent.
- (b) Define the following: [3 × 1 = 3]
 - (i) Vector
 - (ii) Epimer
 - (iii) Helicase.
- (c) Draw the structure of the following: [2 × 1 = 2]
 - (i) A fatty acid with 18 : 3 ($\Delta^{9,12,15}$) composition
 - (ii) 2'-deoxycytidine.

Question 2. Elucidate the following in diagrammatic form only: [2 × 6 = 12]

- (a) Activation energy-based concept of enzyme catalysis
- (b) Four-step events of chain elongation during protein synthesis in prokaryotes.

Question 3. Answer the following:

- (a) Discuss the characteristic properties of peptide bond. [5]
- (b) Describe the transcription initiation assembly in prokaryotes. [7]

Question 4. Answer the following:

- (a) Discuss the different types of intra-molecular interaction which stabilizes 3D structure of a protein. [6]
- (b) What are disaccharides? Describe the structures and functions of any three important disaccharides. [6]

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